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Catastrophic causes of truck drivers' crashes on Brazilian highways: Mixed method analyses and crash prediction using machine learning



Rodrigo Duarte Soliani^a, Ana Rita Tiradentes Terra Argoud^b, Fábio Santiago^c,
Alisson Vinicius Brito Lopes^a, Nwabueze Emekwuru^{d,*}

^a Federal Institute of Acre (IFAC), Av. Brazil, 920, 69.903-06, Rio Branco, AC, Brazil

^b Federal University of São Carlos (UFSCar), Rod. Washington Luis, km 235, 13565-905, São Carlos, SP, Brazil

^c Aeronautics Institute of Technology (ITA), Praça Marechal Eduardo Gomes, 50, 12228-900, São José dos Campos, SP, Brazil

^d School of Mechanical Engineering, Coventry University, Coventry, CV1 5FB. UK

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ABSTRACT

Traffic crashes represent a global challenge, especially in Brazil, where one-third of incidents on federal highways involve trucks, highlighting significant economic and safety risks for truck drivers and the community at large. This study focuses on understanding the specific causes of crashes involving trucks on Brazilian highways, using a decade of data from the Federal Highway Police to develop a predictive model aimed at accident prevention. It analyzes historical crash trends, selects attributes for prediction models, trains classifiers, evaluates predictions through confusion matrices, and enhances reliability via cross-validation techniques, aiming to develop an accident prevention tool. The analysis revealed a temporal pattern, with a slowdown in fatal incidents from 2013 to 2016, followed by an upward trend from 2017. MG-381 emerged as the deadliest highway, and single-lane roads were identified as more accident-prone, emphasizing the need for targeted preventive measures. Additionally, machine learning models achieved an accuracy of over 70 %, with XGBoost and LightGBM leading at 73 %, providing reliable insights for road safety interventions. In transportation engineering and road safety research, these findings highlight the importance of data-driven approaches to understand accident dynamics and design effective interventions to mitigate risks on highways, thereby contributing to increased road safety and social well-being.

Introduction

Traffic crashes pose a global challenge, resulting in loss of human lives, injuries, and significant financial burdens on society. According to the World Health Organization (WHO, 2018), road traffic crashes are the leading cause of death among children and young people aged 5 to 29 years, claiming approximately 1.3 million lives annually. Additionally, 20 to 50 million people suffer non-fatal injuries, many of whom develop disabilities as a result. The WHO report indicates that 93 % of fatalities occur in low- and middle-income countries, despite these nations accounting for only about 60 % of the global vehicle fleet.

* Corresponding author.

E-mail addresses: rodrigo.soliani@ifac.edu.br (R.D. Soliani), ana.terra.argoud@gmail.com (A.R.T.T. Argoud), fabio.santiago737@gmail.com (F. Santiago), britoloo@gmail.com (A.V.B. Lopes), nwabueze.emekwuru@coventry.ac.uk (N. Emekwuru).

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In Brazil, the situation mirrors this global crisis. Between 2007 and 2018, federal highways recorded a total of 570,029 crashes involving at least one truck, according to the National Confederation of Transport (CNT, 2019a). These incidents constituted 33.1 % of all crashes during that period. Specifically, there were 187,489 crashes with victims on federal highways, accounting for 24.8 % of all such incidents nationwide.

Injuries from road traffic crashes impose significant financial costs on individuals, families, and nations. These costs include medical expenses and loss of productivity for those with severe or partial injuries, as well as for family members who must take time off work or school to care for the injured (Zakeri et al., 2021). A study by the Institute of Applied Economic Research (Ipea) and the National Association of Public Transport (ANTP) highlighted that Brazilian society faces an annual burden of approximately R\$ 50 billion due to traffic crashes, with significant costs associated with loss of productivity and hospital expenses (De Carvalho, 2020).

The Brazilian economy relies heavily on road transportation to move its production (Araújo et al., 2024), with various sectors depending almost exclusively on truck freight transport, covering a wide range of products, including essential goods and inputs such as fuels, food, medicines, water, and gas. The profession of truck driver plays a fundamental role in the Brazilian economy, accounting for more than 60 % of all cargo in circulation in the country (Larranaga et al., 2017; Paixao Casaca et al., 2017; Soliani, 2022), which represents a significant portion of the Gross Domestic Product (GDP). In 2021, according to data from the National Association of Cargo Transport and Logistics (NTC, 2022), road transportation moved about 1.1 trillion Ton-Kilometers (TKUs), representing an increase of nearly 10 % compared to the previous year. This expansion is mainly due to three factors: the solid performance of agribusiness (De Oliveira & Alvim, 2017); the continued expansion of agricultural frontiers (Bühler et al., 2016); and the growth of e-commerce and the internalization of production, resulting in longer distances traveled (Arbache, 2015).

Due to Brazil's considerable territorial extent, truck drivers commonly face long working hours, often exceeding 10 hours per day (Giroto et al., 2019). This scenario exposes them to a series of risks. In addition to the distance, there is pressure to meet deadlines, and exhausting workdays lead to a disruption of truck drivers' sleep cycle, resulting in physical, cognitive, and psychological overload (Soliani et al., 2024). This fatigue, coupled with tiredness and physical wear and tear, leads to reflex loss, and compromises the safety of the driver and everyone on the road (Bhuiyan et al., 2022; Chowdhury et al., 2018; Soliani et al., 2023).

The increased risk of crashes is also linked to poor infrastructure conditions, road geometry issues, inadequate signage, and pavement conditions (Martins & Garcez, 2021; Mitra & Bhowmick, 2020; Montoya-Alcaraz et al., 2020). Many roads were not designed to accommodate the growing truck traffic, as evidenced by the low density of paved roads in Brazil. In 2019, only 213,452.8 km (12.4 %) of the country's 1,720,700.3 km of roads were paved (CNT, 2019b).

Given the challenging scenario of traffic crashes in Brazil, particularly on federal highways, it is essential to deepen the understanding of the causes of these crashes, with a special focus on incidents involving trucks. The heavy reliance of the Brazilian economy on road transportation and the high incidence of truck-related crashes underscore the urgent need for studies that can help mitigate these risks.

The central objective of this study is to identify the main underlying causes of crashes on Brazilian federal highways and develop a predictive model to effectively contribute to crash prevention. This research addresses a critical issue impacting public safety and economic efficiency in Brazil. By identifying the causes of truck-related crashes and developing tools to mitigate them, the study aims to reduce fatalities, injuries, and economic losses associated with these incidents.

To achieve this, the study analyzes a decade-long historical series (2013-2022) of traffic crash data recorded by the Federal Highway Police (PRF). Given Brazil's vast territory, the research focuses on the states of São Paulo, Rio de Janeiro, Minas Gerais, Paraná, and Rio Grande do Sul, which together accounted for 64 % of the country's GDP from 2013 to 2022 (IBGE, 2023). Through time series analysis and the use of classifier algorithms, the study investigated the primary causes of crashes, considering variables such as traffic conditions, geographical location, weather conditions, and vehicle characteristics. The goal is to provide a practical predictive tool for preventing these crashes, thereby enhancing road safety, and improving the quality of life for society.

Theoretical framework

Causality of truck crashes

The causality of crashes can be examined through various perspectives. Generally, the analyzed factors are divided into different categories, including: characteristics of road transport companies, driver aspects, external road attributes, and vehicle-related factors (Ashraf et al., 2019).

Through a truck safety analysis using data from the Federal Motor Carrier Safety Administration (FMCSA), the regulatory agency for road transport in the United States, Zheng et al. (2018) identified the significant impact of company and driver characteristics on the severity of injuries in truck crashes. The results corroborate previous findings, indicating that conditions such as wet road surface, poor visibility, and crosswinds increase the risk of serious crashes. Young and elderly drivers are identified as groups most susceptible to fatal crashes. An intriguing observation is that fatal crashes occur more frequently in favorable weather conditions, possibly due to decreased vigilance. Additionally, the study highlights the impact of company and driver characteristics, showing that carriers registered in certain states and with specific safety inspection values are more likely to have severe outcomes. Companies with 2 to 5 trucks are associated with the lowest risk of crashes, while companies with higher inspection values indicate a less satisfactory safety record.

Lemke et al. (2021) investigated the factors associated with compliance with the working hours of 262 truck drivers through a cross-sectional observational approach, evaluating its impact on safety risk. The results emphasized that higher scores in the variable composed of working hours were directly linked to higher weekly mileage, prolonged working hours, accelerated work pace, and

lower sleep quality. In the analysis model employed by the authors, statistically significant predictors were identified, such as traveling less than 4,000 miles (6,400 km) per week, working fewer than 11 hours per day or between 11 and 13 hours per day, and a lower frequency of accelerated work pace.

The study conducted by [Ladeira et al. \(2017\)](#) analyzed the global burden of disease from traffic crashes in Brazil from 1990 to 2015, using data from the 2015 Global Burden of Disease study. Brazil is identified as the second South American country with the highest Disability-Adjusted Life Years (DALYs) rate due to traffic crashes, behind only Paraguay. Surprisingly, DALY rates in Brazil surpass those of countries with lower socioeconomic indicators, such as Ecuador and Bolivia. A comparative analysis with BRICS countries reveals relatively similar DALY rates related to traffic crashes, except for South Africa, which records a higher rate. In Brazil, China, and India, pedestrian injuries are the main component of DALYs, while in Russia and South Africa, the highest rates are attributed to injuries to car occupants. The study emphasizes the urgent need for adequate investments in road safety systems in the face of rapid economic growth and increased motor vehicle traffic in BRICS countries, aiming to reduce injuries and deaths.

[Mashhadi et al. \(2018\)](#) uncovered distinct factors contributing to crashes, distinguishing between those involving a single vehicle and those involving multiple vehicles. The first approach investigated the factors that increased the chances of injuries or deaths in crashes involving only one truck and in crashes involving multiple vehicles, with at least one truck. The analysis revealed that being female, driving on dry roads, exceeding speed limits (above or below), experiencing rollovers or steering failures, and having distractions in the cabin increased the chances of injuries or deaths in crashes involving only one truck. In crashes involving multiple vehicles with trucks, not complying with speed limits, driving on segments with speed limits beyond what is allowed, and driving on weekends increased the chances of injuries or deaths.

In their research, [Yuan et al. \(2021\)](#) aimed to investigate the determinants of fatal collisions involving trucks, focusing on truck drivers. The analysis was based on demographic characteristics, criminal history, and infringement behaviors. Three distinct groups of drivers were identified: the first group consisted of middle-aged and elderly drivers with low risk of violations and crash history; the second group comprised drivers with a high risk of violations and crash history; and the third group was formed by middle-aged drivers without violations. For the high-risk group, associated with risky behaviors such as driving under the influence of alcohol or drugs, specific safety measures such as advanced information systems and stricter penalties are recommended. On the other hand, the low-risk group exhibits better driving habits, while the third group was identified as the safest. Additionally, factors such as adverse weather, rural areas, and collision types influenced the severity of crashes in all groups.

[Ahmed et al. \(2018\)](#) analyzed the impact of truck involvement on the severity of crashes in Wyoming (USA), a predominantly rural state, over a 10-year period (2002-2011). The analysis revealed that heavy trucks accounted for 68 % of vehicles involved in crashes on interstate highways. Additionally, crashes involving trucks occurred 19 % more frequently in adverse weather conditions, such as strong winds or snow, compared to crashes without truck involvement. The analysis extended to verifying whether the involvement of a heavy truck is a significant predictor of crash severity and identifying road and weather factors that affect this severity. Bayesian binary logistic models were calibrated to verify the results of the statewide data analysis and to study the relationship between truck involvement and crash severity. The results show that truck involvement in crashes results in greater severity, with the chances of a serious crash increasing by 2.3 and 4.5 times on state and interstate highways, respectively.

The study by [Eboli et al. \(2020\)](#) highlights the significant influence of road-related factors on the nature of crashes, including road type, road configuration, location, and signage. For less severe crashes, age, time of day, day of the week, and vehicle type are relevant, while gender, road surface, and license type are not significant. In severe crashes, elements such as road segment, road type, road surface, time of day, license type, and possession of a driver's license exert influence, while gender and weather conditions have no impact. Regardless of the crash severity, improvements in road-related factors, such as intersection and urban road design, are crucial for road safety, including the promotion of one-way roads. Reducing less severe crashes requires traffic condition improvements, especially at night, while strict control of driver-related factors is crucial to mitigate severe crashes, mainly among less experienced drivers.

Studies on traffic crashes indicate a significant increase, especially in urban areas, where fatalities and non-fatal injuries involving cargo vehicles have increased more rapidly than overall traffic crash rates. According to [McDonald et al. \(2019\)](#), the growth of e-commerce has challenged urban road safety, increasing cargo flow on roads and avenues. Rates of non-fatal injuries in cargo vehicle crashes in these areas increased significantly between 2009 and 2015, mainly during weekdays and peak cargo demand hours. These changes highlight the need for enhanced surveillance systems to capture cargo vehicle crashes and comprehensive urban planning, considering cargo distribution patterns.

In China, despite the reduction in the average traffic fatalities per capita, fatality and injury rates continued to rise. The study by [Wang et al. \(2019\)](#) analyzed the severity of traffic crashes from 2000 to 2016, using two indicators: human damage (HD) and case fatality rate (CFR). During this period, there was a 19.0 % increase in the overall CFR and a 63.7 % increase in HD. Crashes involving cargo vehicles were more severe and had a higher risk of colliding with smaller cars. The speeding problem also worsened, with a CFR 42.0 % above the average in 2016. These trends contrast with the overall decrease in traffic crashes, highlighting the importance of dissipating impact energy and victim protection measures in the severity of traffic crashes.

The results of [De Oliveira and Achcar's \(2020\)](#) study reveal the grim reality in Brazil, showing high rates of victims with severe or fatal injuries in traffic crashes. In January 2019, out of 11,809 injured on Brazilian federal highways, 5,201 people (44.04 % of the total) were uninjured, 4,837 (40.06 % of the total) had minor injuries, 1,394 (11.80 % of the total) suffered severe injuries, and 377 (3.19 % of the total) resulted in deaths. The results emphasize the need to strengthen surveillance on federal highways to prevent crashes caused by animals on the road, improve signage, and possibly implement more devices to prevent improper overtaking, as well as intensify driver monitoring, especially during the early morning, when the risk of severe injuries is considerably higher due to drowsiness.

According to [De Oliveira and Achcar \(2020\)](#), the presence of single-lane highways and crashes involving motorcyclists are significant factors for the number of victims with severe or fatal injuries. Statistical analysis revealed a statistical relationship between the severity of victims' injuries and various factors, including the cause of the crash, type of crash, day of the week, type of highway, and type of vehicle involved. Factors such as age, time of day, road conditions, gender, and type of vehicle also proved significant for fatal victims, highlighting alarming numbers of deaths, often involving young individuals in the prime of their lives. These results reinforce the need for greater surveillance on federal highways to address issues such as the presence of animals on the road, improper overtaking, and monitoring of driver conditions, especially regarding fatigue, which can significantly increase the number of severe injuries, especially during dawn. Other significant factors that significantly increase the number of victims with severe injuries include single-lane highways and crashes involving motorcyclists.

The study by [Xu et al. \(2020\)](#) conducted an analysis of the distribution of extremely severe crashes (with more than 10 fatalities) regarding gender, driving experience, road technical level, road alignment, time of occurrence, and weather conditions. The results indicated that the majority of extremely severe crashes were caused by male drivers (97.8 %), with 6 to 10 years of driving experience (23.7 %), mainly on federal highways (42.8 %).

[Xu et al.'s \(2020\)](#) analysis revealed significant differences between extremely severe crashes (with more than 10 deaths) and normal crashes, highlighting that these crashes are more likely to occur on federal highways, especially on segments with poor road conditions (horizontal curves with a downward slope) and in adverse weather conditions, such as rain and fog. The Fault Tree Analysis (FTA) identified multiple combinations of factors that can lead to extremely severe crashes, grouped into minimal sets, representing the smallest combinations of basic events that can result in severe crashes. For example, heavy vehicle overload, when associated with braking problems or lack of nighttime lighting, was identified as an important factor. Excessive speed, especially in adverse weather conditions, and vehicle mechanical problems also contributed to these crashes.

Although the previous research, as reviewed in the preceding paragraphs, provides significant understanding into the factors contributing to truck crashes, limitations to the understanding also exist. Many of the studies focused on specific aspects of crashes, such as driver characteristics or road conditions, without fully considering the complexity of the interactions between these factors. Additionally, there is a dearth of studies that address specific regional contexts, which may limit the generalizability of the findings.

The study presented in this paper overcame these limitations by adopting a more comprehensive and integrative approach. By analyzing a broad spectrum of factors—driver characteristics, road conditions, regional context, and their interactions—a deeper understanding of truck crashes were achieved. Also, by focusing on areas underrepresented in previous research, such as regional contexts and inter-factor relationships, this paper fills critical gaps in the literature. Consequently, the present study not only advances academic knowledge about truck crashes but also has the potential to make a significant impact on the prevention and mitigation of these events.

Methodology

This research adopts an applied approach, aiming to provide practical insights and solutions to a specific problem ([Gil, 2017](#)). Its objectives, in turn, encompass both an exploratory and descriptive perspective ([Mohajan, 2020](#)). While exploration seeks a deep understanding of the dynamics of crashes involving truck drivers on federal highways, the descriptive approach meticulously details the factors contributing to this phenomenon. To achieve these objectives, this research adopts a quantitative approach, using robust methods of data collection and analysis, as well as advanced preprocessing techniques, culminating in the use of machine learning algorithms.

The study initially focuses on truck drivers traveling on federal highways in the states of São Paulo, Minas Gerais, Rio Grande do Sul, Rio de Janeiro, and Paraná. These states were strategically selected as they represent a significant portion, approximately 62.8 %, of the Brazilian Gross Domestic Product (GDP), as indicated by [Carneiro \(2023\)](#).

Drawing on references such as [Harrison \(2019\)](#), [Géron \(2019\)](#), and [Grus \(2021\)](#), the authors outline a sequence of steps to conduct the research: from the initial data collection, through preprocessing of information, analysis of historical series, and identification of crashes causes, to the selection and training of machine learning classifier models. Finally, the stage of prediction and evaluation of the results obtained is proposed, which will be thoroughly explained below.

Data collection

To meet the objectives of this research, the authors chose to use data provided by the Traffic Accident Bulletin (BAT) of the Federal Highway Police (PRF), an institutional source linked to the federal government, aiming to reinforce the solidity and reliability of the study scope. Data for the decade between 2013 and 2022 were collected from this source and processed using the Python language.

The numbers of incidents recorded by the PRF each year are presented in [Table 1](#), allowing for a macro understanding of the evolution of the number of records throughout the decade, highlighting the year 2021 with the lowest number of records in the historical series and the year 2022 with the highest number of registered occurrences.

The total number of records encompasses all occurrences registered by the PRF over the decade under analysis. It is important to note that these numbers include not only Traffic Accident Bulletin involving trucks but also all other types of vehicles circulating on federal highways. Given this breadth, it is crucial to select the target audience of interest, namely, truck drivers from the five states that contribute the most to the national GDP, which will be done during the detailed preprocessing phase to follow.

Table 1
Traffic crash records by Brazilian PRF (2013-2022).

Year	Number of crash records
2013	405.817,00
2014	368.506,00
2015	269.052,00
2016	216.259,00
2017	204.405,00
2018	164.887,00
2019	162.494,00
2020	394.735,00
2021	150.902,00
2022	519.023,00
Total	2.856.080,00

Source: Adapted from [PRF \(2024\)](#).

Data preprocessing

The data preprocessing stage aims to enable the analyses and implementations to be carried out in this study. Initially, the databases for each year were concatenated, resulting in a single database containing 2,856,080 records. Next, the target audience for this study was selected, consisting of truck drivers from the states of São Paulo, Rio de Janeiro, Minas Gerais, Rio Grande do Sul, and Paraná, totaling 207,665 records. With the target audience defined, the data cleaning process began, as described by [Harrison \(2019\)](#), [Géron \(2019\)](#), and [Grus \(2021\)](#). This process involved removing missing values, duplicate data, corrupted data, among others. At the end of this process, a total of 190,397 records were intact and ready to be used in this study.

After selecting the target audience and cleaning the database, the following attributes were chosen for each record in the database: date, UF (state), cause of the crash, type of crash, time of day, weather condition, type of vehicle, type of road, day of the week, time, cause of the crash, highway, driver's age, and vehicle manufacturing year. It is important to highlight that these attributes were present in all years of the database, allowing for analysis over time. Furthermore, this attribute selection step did not alter the number of records to be used in the study; therefore, the resulting quantity from this step is the same as before, namely, 190,397 records.

Understanding historical series and original causes of crashes

By chronologically ordering the databases based on the dates of the crashes, we can obtain the historical series of the numbers of fatal crashes on the highways under study, as demonstrated in [Fig. 2](#). To understand the circumstances in which the crashes occurred, we identified the highway on which the vehicle was (see [Fig. 3](#)), the type of road (see [Fig. 4](#)), the type of crash (see [Fig. 5](#)), and the influence of the time of day and weather conditions on the number of recorded crashes (see [Fig. 6](#)), enabling a robust analysis of the circumstances of the crashes.

Selection of classifier machine learning models

Among the various classification algorithms available in the literature, we opted for those considered classical and available in machine learning libraries in Python, such as Scikit-learn. Thus, we chose: GaussianNB (GNB), K-Nearest Neighbor (KNN), Decision Tree (DT), Random Forest (RF), Logistic Regression (LR), Light Gradient Boosting Machine (LGBM), DMLC XGBoost, and Multilayer Perceptron (MLP).

Training of classification algorithms

After selecting the classifier algorithms, it becomes essential to train these models for their subsequent application in predictions. The process begins with the careful selection of predictive attributes and classes. For predictive attributes, we considered the cause of the crash, type of crash, highway on which the vehicle was, type of road, weather condition, time of day, type of vehicle, driver's age, and vehicle manufacturing year. Regarding the target classes, we distinguished whether the crash resulted in fatalities or not.

Next, we divided the database into training and testing sets, following the proportions recommended by [Xu and Goodacre \(2018\)](#): 75 % of the data for training, equivalent to 133,277 records, and 25 % for testing, totaling 57,120 records. With the predictive attributes, classes, and correct data division in hand, we proceeded to select the hyperparameters to optimize the accuracy, precision, and recall metrics, detailed in [Table 2](#) below.

After defining the hyperparameters, the algorithms are ready to be trained and then make predictions.

Prediction and evaluation of algorithms

The predictions made by the algorithms are carried out using the test dataset, composed of 57,120 records with their predictive attributes. Each algorithm is applied to these records to classify them regarding to the occurrence of crashes with victims. Then, the

Table 2
Hyperparameters of machine learning algorithms.

Classifier	Hyperparameters
GaussianNB	<i>None</i>
K-Nearest Neighbor	<i>n_neighbors</i> =30, <i>metric</i> = “minkowski”
Decision Tree	<i>criterion</i> = “entropy”, <i>max_depth</i> =15
Random Forest	<i>n_estimators</i> = 150, <i>criterion</i> = “gini”, <i>random_state</i> =0, <i>max_depth</i> =15
Logistic Regression	<i>'num_leaves':</i> 150, <i>'objective':</i> “binary”, <i>'max_depth':</i> 20, <i>'learning_rate':</i> 0.1, <i>'max_bin':</i> 20
LighGBM	<i>max_depth</i> =20, <i>n_estimators</i> =150, <i>random_state</i> =3, <i>learning_rate</i> =0.2
XGBoost	<i>max_iter</i> =1000, <i>penalty</i> =‘l2’, <i>solver</i> = ‘lbfgs’
MLP	<i>max_iter</i> =1000, <i>hidden_layer_sizes</i> = (100)

Table 3
Metrics used in the analysis.

Metrics	Expression
Accuracy	$(TP+TN)/(TP+FN+FP+TN)$
Precision	$TP/(TP+FP)$
Recall	$TP/(TP+FN)$

True Label	True Positive (TP)	False Negative (FN)
	False Positive (FP)	True Negative (TN)
Predicted Label		

Fig. 1. Confusion Matrix.

predictions are evaluated using the confusion matrix, presented in Fig. 1, along with the associated metrics, as shown in Table 3. For this classification algorithm evaluation, the `Sklearn.metrics.confusion_matrix` library available in Scikit-learn was used.

Fig. 1 illustrates the confusion matrix, where correctly classified records by the algorithms are highlighted in the True Positive (TP) and True Negative (TN) fields, as well as the errors made, evidenced by the False Negative (FN) and False Positive (FP) fields.

In addition to the values mentioned earlier, we can calculate accuracy, precision, and recall metrics from them, as shown in Table 3, which helps us better understand the performance of the classifier algorithms.

The results obtained from the metrics listed in the previous table play a fundamental role in evaluating the correct choice of hyperparameters and predictive attributes, as they should be selected to optimize accuracy, precision, and recall values. This analysis is essential to ensure that the models surpass the randomness of the phenomenon, thus demonstrating significant learning by the algorithms.

Despite the robustness of the confusion matrix in measuring the performance of classification algorithms, it is important to note that it is unable to assess the generalization of these models, as presented in [Sklearn.org](https://sklearn.org). To address this issue, as described in [Sklearn.org](https://sklearn.org), it is necessary to apply the KFold technique, also known as cross-validation. This technique involves dividing the dataset into *k* mutually exclusive and equally sized subsets. One subset is used for testing while the other *k*-1 subsets are used for training. In this work, *k*=10 was chosen to ensure adequate computational time and a sufficient amount of data for training. This process is repeated *k* times, with cyclic alteration of the test set in each iteration, calculating the model's accuracy each time.

With the implementation of these two levels of evaluation, along with the variety of algorithms used in the classifications, it is possible to achieve a high and rigorous level of reliability in the predictions made.

Results and discussion

The analysis of the scenario of crashes involving truck drivers on Brazilian federal highways from 2013 to 2022 begins with the quantitative survey of mortality time series, as illustrated in Fig. 2 and detailed in the following Table 4. The combined analysis of these data visualization tools enables us to understand the dynamic behavior of the number of deaths over this period.

It was observed that 2022 was the deadliest year, registering 1328 deaths, with an average monthly rate of 110.67, with the months of January, February, April, May, July, and September being particularly critical, with values above the annual average. In second place, 2020 was the second deadliest year, with 1237 deaths, presenting an average monthly rate of 103.08. Noteworthy are the months of January, July, August, September, October, and December, which recorded values above the annual average. These findings underscore the gravity of the situation and the urgency for intervention measures.

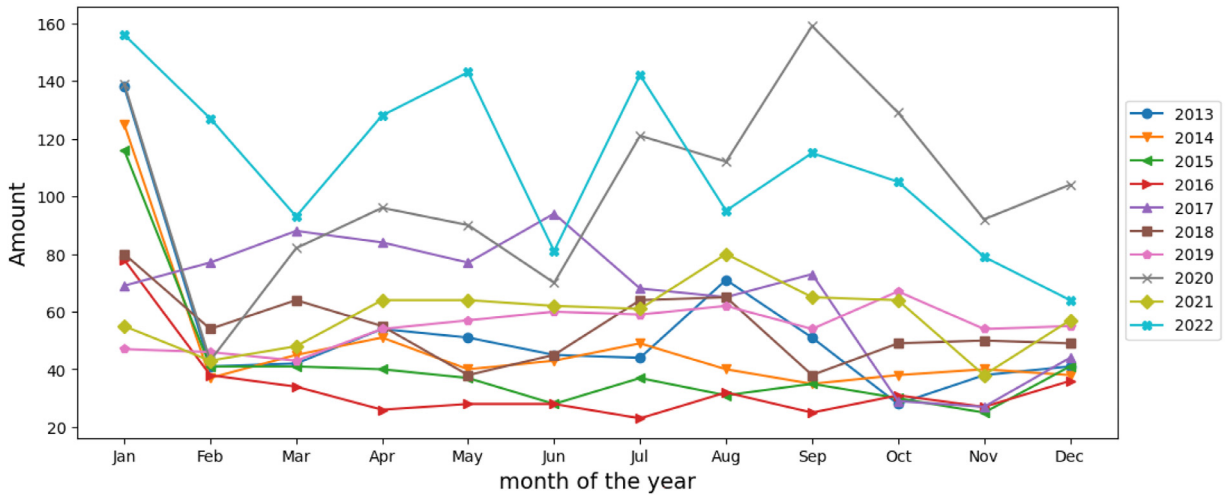


Fig. 2. Annual variation in the number of deaths involving truck drivers on Brazilian federal highways (2013–2022).

Table 4

Monthly death toll involving truck drivers on Brazilian federal highways (2013–2022).

Year Month	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
	Number of deaths registered									
January	138	125	116	78	69	80	47	139	55	156
February	41	37	41	38	77	54	46	43	43	127
March	42	45	41	34	88	64	43	82	48	93
April	54	51	40	26	84	55	54	96	64	128
May	51	40	37	28	77	38	57	90	64	143
June	45	43	28	28	94	45	60	70	62	81
July	44	49	37	23	68	64	59	121	61	142
August	71	40	31	32	65	65	62	112	80	95
September	51	35	35	25	73	38	54	159	65	115
October	28	38	30	31	29	49	67	129	64	105
November	38	40	25	27	27	50	54	92	38	79
December	41	38	41	36	44	49	55	104	57	64
Total	644	581	502	406	795	651	658	1237	701	1328
Average	53,67	48,42	41,83	33,83	66,25	54,25	54,83	103,08	58,42	110,67
SD	27,32	23,55	22,97	14,01	20,96	11,75	6,64	30,25	10,8	27,87
Variation (%)	-	-9,78	-13,6	-19,12	95,81	-18,11	1,08	87,99	-43,33	89,44

In the intermediate range, with death tolls ranging between 600 and 800, the years 2013, 2017, 2018, 2019, and 2021 stand out, with 644, 795, 651, 658, and 781 fatal crashes, respectively. Notably, the years 2017 and 2019 present monthly averages of deaths involving truck drivers in the ranges of 54.83 and 58.42 fatalities. In contrast, the years 2014, 2015, and 2016 record the lowest levels of crashes, with values of 581, 502, and 406 deaths, corresponding to monthly averages of 42, 41.83, and 33.63, respectively.

When analyzing the chronology of crashes throughout the decade from 2013 to 2022, a deceleration in fatal incidents is observed between 2013 and 2016. However, in 2017, there was an increase of 95.81 % in crashes involving truck drivers compared to the previous year. This trend remained almost constant between 2018 and 2019, with approximately 650 deaths annually, but rose again in 2020, oscillating in the following years with an upward trend. Therefore, there are two distinct moments in the study period: from 2013 to 2016, with a deceleration in the number of crashes, and from 2017 to 2022, characterized by oscillations with an upward trend.

After examining the dynamics of the number of deaths throughout the decade from 2013 to 2022, this study now turns to a detailed investigation of the circumstances surrounding these crashes, seeking to identify their underlying causes. This initial step involves identifying the ten highways with the highest numbers of deaths, as presented in Fig. 3 below. The state of Minas Gerais stands out in this representation, contributing five of the ten deadliest highways, led by MG-381 in terms of fatal victims. Next are the states of Paraná, with three highways among the ten most dangerous, and Rio de Janeiro and São Paulo, each with one highway on the list. Despite São Paulo having only one listed highway, SP-116 ranks second in the number of deaths among the ten mentioned here.

The MG-381, with a length of 1,181 kilometers, connects São Mateus in Espírito Santo to São Paulo. Recognized as the 'Highway of Death,' the stretch linking the states of Minas Gerais and Espírito Santo is known for its winding layout, contributing to frequent crashes and making it one of the most dangerous road sections in Brazil (Brazil, 2021; Deivielison et al., 2015). In addition to its

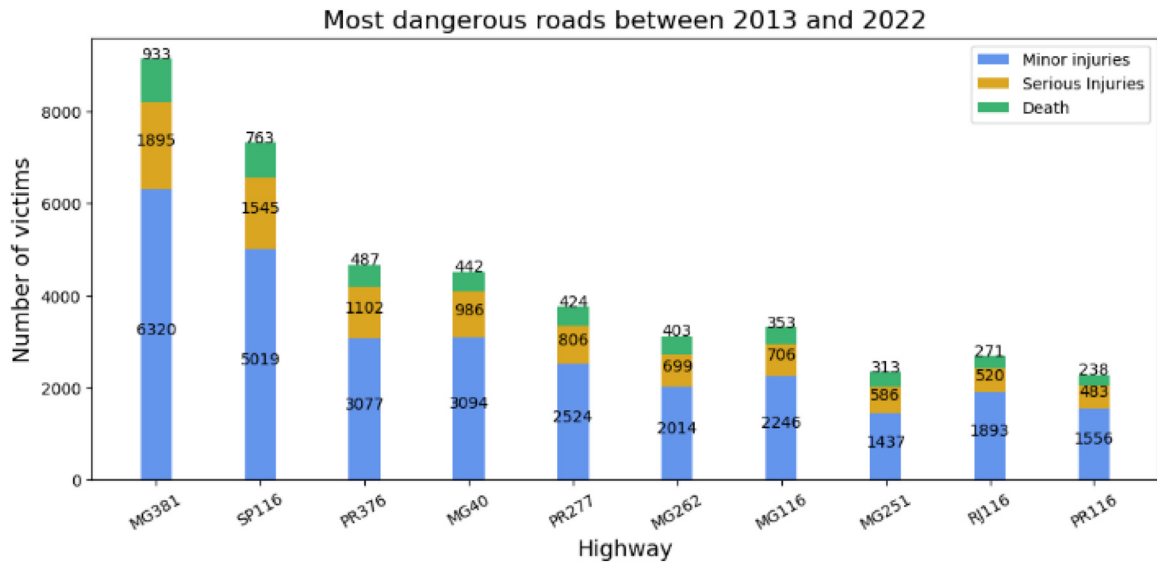


Fig. 3. The 10 Brazilian federal highways with the highest number of fatal crashes involving truck drivers (2013-2022).

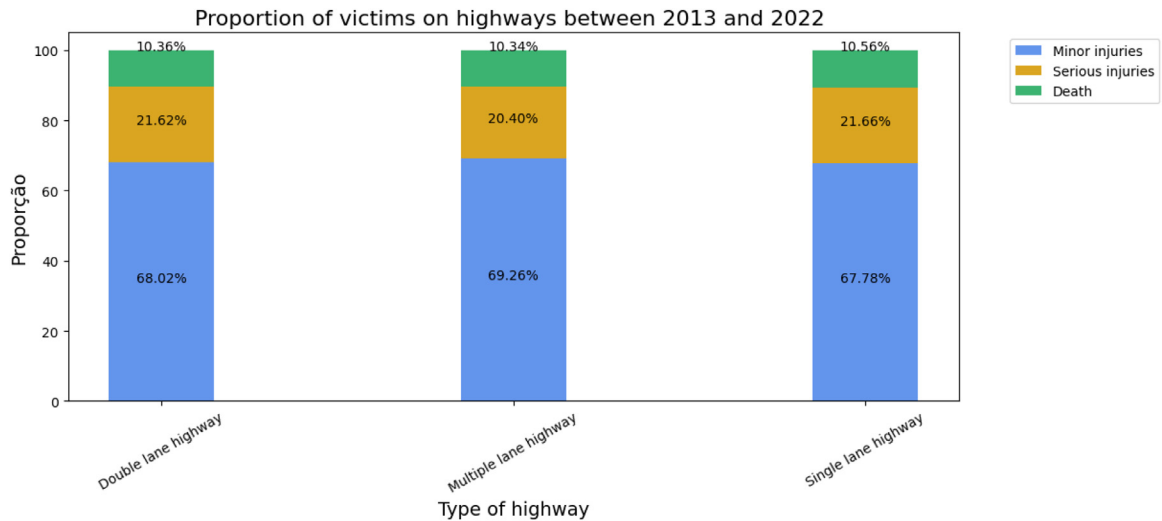


Fig. 4. Distribution of victims by type of highway (2013-2022) in crashes involving truck drivers.

concerning reputation, the MG-381 plays a crucial role in the economic development of the region, passing through important cities such as Governador Valadares, Ipatinga, Coronel Fabriciano, Timóteo, and João Monlevade, and encompassing the largest steelmaking hub in Latin America (Brazil, 2022).

After identifying the highways with the highest fatality rates, the study now investigated the influence of road type on crash occurrence, as presented in Fig. 4.

The analysis reveals that the rates of deaths, serious injuries, and minor injuries vary slightly with each type of road. However, it is noteworthy that highways with dual and multiple lanes exhibit the highest proportions of minor injuries, registering 68.02 % and 69.26 %, respectively, indicating greater safety on these highways. On the other hand, the highest proportion of fatal crashes (10.56 %) occurs on single-lane roads, positioning them as the most dangerous among those analyzed.

The study conducted by Haghighi et al. (2018) used data collected from 826 segments of multi-lane highways in the state of Indiana, USA, covering vehicle crashes recorded over a three-year period, along with information on traffic patterns, road geometry, and pavement surface conditions. Variables considered significant in relation to crashes include exposure-related measures (such as segment length), geometric characteristics (such as lane width, left shoulder width, median width, and tangent segments), road type (single or multi-lane), and pavement surface condition.

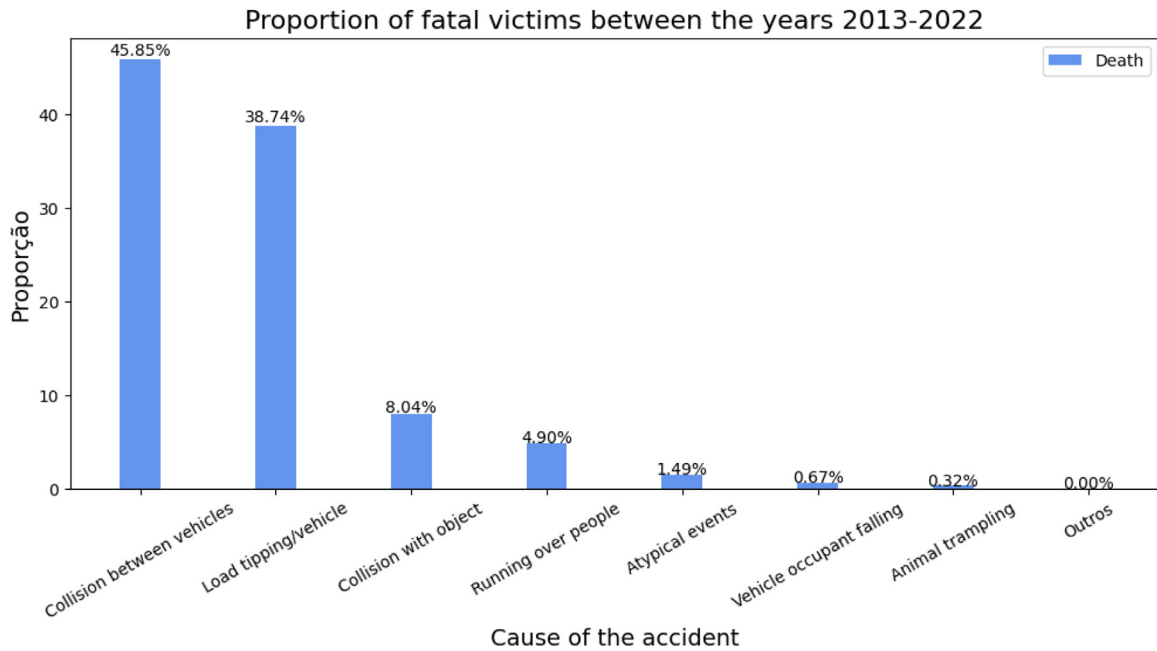


Fig. 5. Analysis of causes of deaths involving truck drivers on Brazilian federal highways (2013-2022).

As mentioned by Zhang et al. (2016), it is common for multi-lane highways to exhibit superior quality and better conditions, resulting in a lower likelihood of drivers causing severe crashes. Furthermore, roads equipped with traffic lights and signage help increase driver attention, leading to a reduction in the likelihood of fatigue-related crashes.

By detailing the number of deaths in relation to road type, it is possible to identify the atypical factors responsible for crash occurrences. As demonstrated in Fig. 5, vehicle collisions are the main cause of deaths, accounting for 45.85 % of incidents. This category includes various types of collisions such as head-on, side, and diagonal. The second most frequent cause of deaths involving truck drivers is cargo rollovers, which consequently lead to vehicle rollovers.

The study also identifies critical factors that contribute to fatal incidents, such as vehicle collisions and cargo rollovers. These findings align with those of Rolison et al. (2018), who investigated the causes of road accidents involving drivers of varying ages and genders. They compared the perspectives of police officers and the general public with actual accident records. The study revealed that both groups accurately recognized factors linked to crashes involving young and older drivers, attributing risk-taking behaviors more frequently to younger drivers and medical conditions to older ones. Greater awareness of certain factors, such as drugs/alcohol and inexperience, was observed among police officers, particularly in scenarios involving young drivers. However, discrepancies emerged between the factors identified by participants and those documented in accident records, particularly regarding the underreporting of distractions like mobile phone use.

The inherent complexity of road traffic safety results from the interaction between the driver, vehicle, road, and environment. This interaction is often considered the main cause of vehicle rollovers. Several studies indicate that most traffic crashes are related to inappropriate driving behaviors, such as speeding, sharp curves, and sudden braking (Ahmed et al., 2022; Bako, 2021; Wang et al., 2023).

The research by Xin et al. (2021) highlights the impact of speed on truck safety during sharp curves, which can compromise the vehicle's trajectory and stability. In road design, it is assumed that the coefficient of lateral force " μ " depends on three variables: speed, radius of the circular curve, and superelevation. It is also assumed that the vehicle strictly adheres to the prescribed path when traveling on the circular curve and performs uniform circular motion (AASHTO, 2018). However, it is important to note that for trucks with worn tires or under adverse road conditions, safety may be compromised, even with the minimum radius of the curve used in road design.

Driving above the speed limit significantly increases the risk of rollover on sharp curves, especially for trucks. The UK's External Vehicle Speed Control (EVSC) project emphasizes that the implementation of a simple and mandatory system in all vehicles, preventing them from exceeding the speed limit, has the potential to reduce injury crashes by 20 % and fatal crashes by 37 % (Gámez Serna & Ruichek, 2017). The advancement of Intelligent Transportation Systems (ITS) has brought innovative approaches to addressing issues related to negotiating curves, such as the Curve Speed Warning (CSW) System, which assesses if a vehicle is approaching curves too quickly.

With the analysis of road type and the main causes of crashes completed, the study now focused on understanding the influence of external factors on crash occurrence. Following this, Fig. 6 presents the mapping of the number of fatal crashes in relation to the time of day and weather conditions, through a heat map.

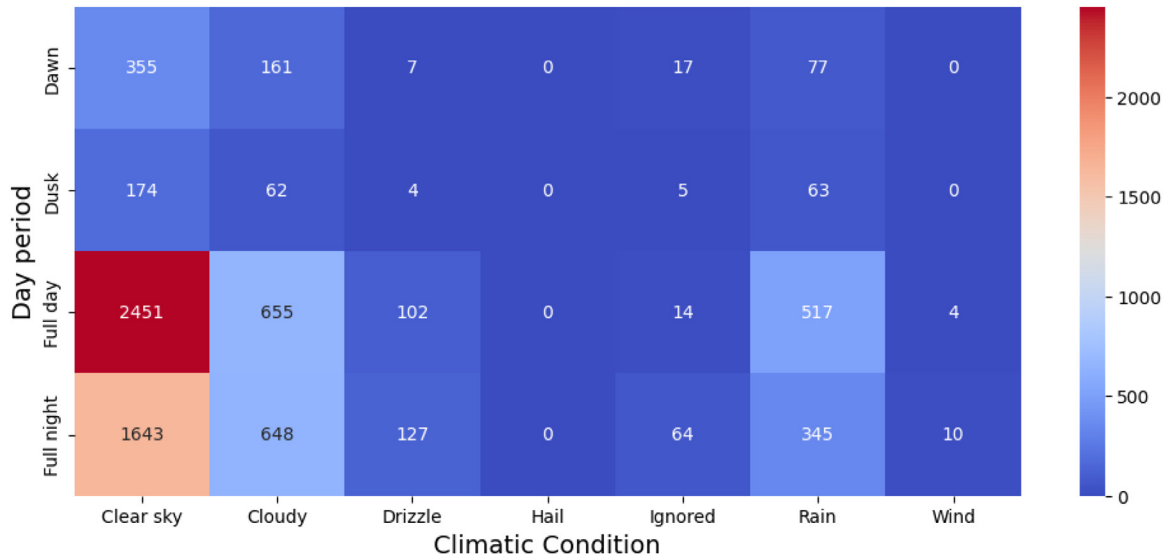


Fig. 6. Influence of weather and time of day on the number of deaths involving truck drivers on Brazilian federal highways (2013-2022).

The analysis reveals that most crashes occur under clear sky conditions, both during the day (2,451 records) and at night (1,643 records). On the other hand, cloudy conditions, during any time of the day, present 1,526 occurrences, approaching the number of crashes under clear sky conditions and during the night. However, it is relevant to note that cloudy conditions often precede rainfall. When we add crashes under rain, regardless of the period, we total 1,002 occurrences. By combining occurrences under cloudy conditions, we reach 2,530 occurrences, the second-highest number, behind only crashes under clear sky conditions, regardless of the time of day, with 4,623 records.

The study conducted by [De Oliveira and Achcar \(2020\)](#), utilizing data from the Brazilian Federal Highway Police spanning all federal highways throughout January 2019, highlighted distinct patterns in injury severity among accident victims based on the time of day and prevailing weather conditions. Specifically, the analysis revealed that the early morning hours exhibit a heightened incidence of severe injuries among accident victims, contrasting with fewer severe injuries during daylight hours. Additionally, certain weather conditions such as wind, clear skies, and cloudy days are correlated with increased rates of severe injuries on federal highways.

The present study expands on the previous studies by undertaking a comprehensive examination of fatal accidents involving truck drivers on Brazilian federal highways from 2013 to 2022, uncovering enduring patterns and pivotal timeframes. Through meticulous scrutiny, the ten most hazardous highways, notably highlighting MG-381, dubbed the 'Highway of Death,' offer crucial insights for targeted interventions. Additionally, the scrutiny of road types identifies single-lane highways as the most perilous, signaling imperative zones for infrastructure enhancements. By dissecting the primary causes of fatalities, such as vehicular collisions and cargo rollovers, and scrutinizing the impact of weather conditions and time of day, a robust groundwork is laid for crafting safety protocols and educational campaigns. Furthermore, juxtaposing the findings with global research data contextualizes Brazil's road safety landscape, while the discourse on cutting-edge technologies like the Curve Speed Warning (CSW) and External Vehicle Speed Control (EVSC) hints at promising advancements in accident prevention.

In the subsequent section, an analysis is conducted of fatal accidents involving truck drivers on Brazilian federal highways between 2013 and 2022. Utilizing both quantitative and qualitative studies, the collected information is explored for the implementation of machine learning models selected during the investigation. Employing both the Confusion Matrix and the KFOLD technique, the accuracy of the classifiers is assessed, providing insights into the patterns identified in the accidents, as well as the predictive capacity of the machine learning algorithms.

Crash prediction

After conducting both quantitative and qualitative studies on crashes involving truck drivers, this research leverages the gathered information to implement the machine learning models selected during the investigation. The Confusion Matrix for each of the algorithms listed in [Table 2](#) is now depicted in [Fig. 7](#) below.

Based on the Confusion Matrix presented, we can evaluate the metrics defined in [Table 5](#), whose calculations will be detailed below.

A preliminary analysis of the applied models reveals that all surpass the 59 % rate, which consists of the probability of random success in the occurrence of a truck-related crash, considering the data from this research. This is evidenced by examining the Precision metric for the "Yes" class (with victims), in which all algorithms record values above 56 %, with [LightGBM](#) and [XGBoost](#)

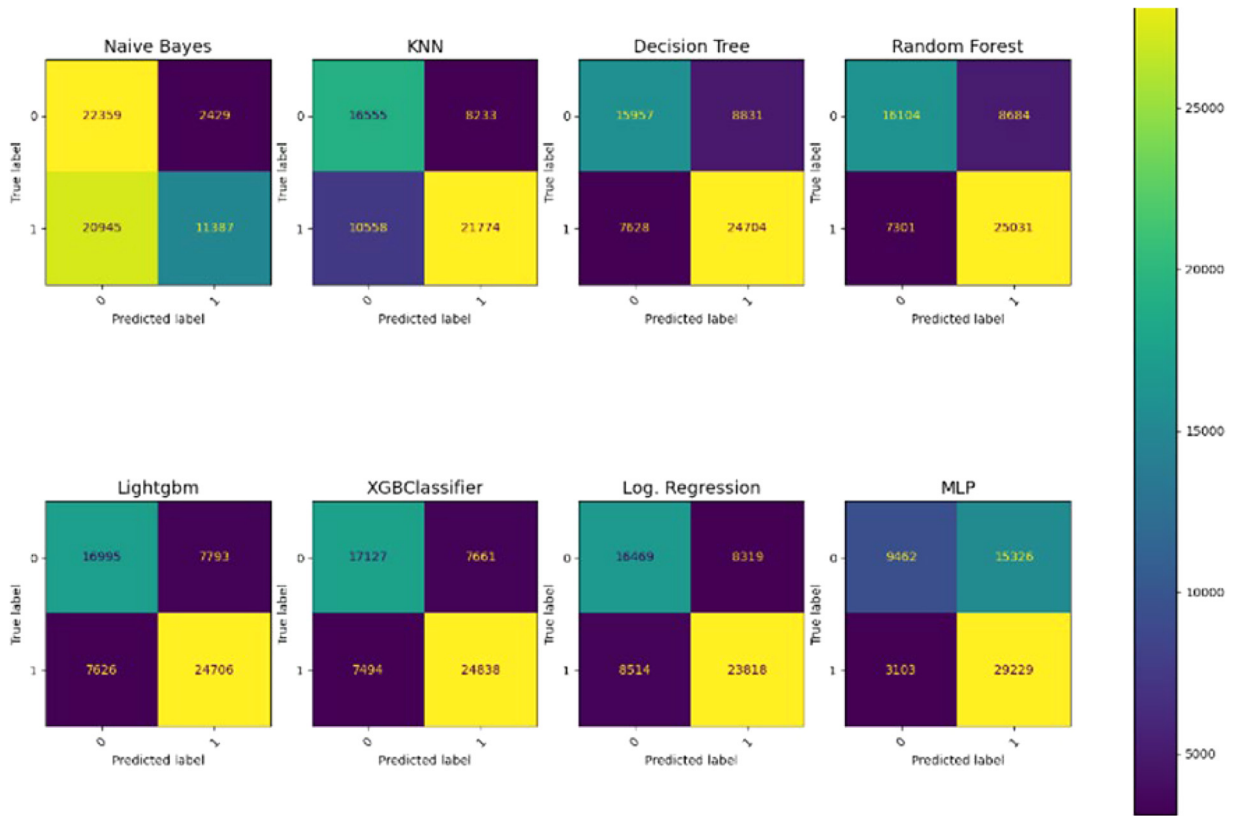


Fig. 7. Confusion Matrix of the classifier algorithms.

Table 5

Metrics associated with classification matrices.

Classifier	Classes		Metrics		
			Precision	Recall	F1_score
Naive Bayes	Victims	No	0.52	0.90	0.66
		Yes	0.82	0.35	0.49
		Accuracy			0.59
KNN	Victims	No	0.61	0.67	0.64
		Yes	0.73	0.67	0.70
		Accuracy			0.67
Decision Tree	Victims	No	0.68	0.64	0.66
		Yes	0.74	0.76	0.75
		Accuracy			0.71
Random Forest	Victims	No	0.69	0.65	0.67
		Yes	0.74	0.77	0.76
		Accuracy			0.72
LightGBM	Victims	No	0.69	0.69	0.69
		Yes	0.76	0.76	0.76
		Accuracy			0.73
XGBoost	Victims	No	0.70	0.69	0.69
		Yes	0.76	0.77	0.77
		Accuracy			0.73
Logistic Regression	Victims	No	0.66	0.66	0.66
		Yes	0.74	0.74	0.74
		Accuracy			0.71
MLP	Victims	No	0.75	0.38	0.51
		Yes	0.66	0.90	0.76
		Accuracy			0.68

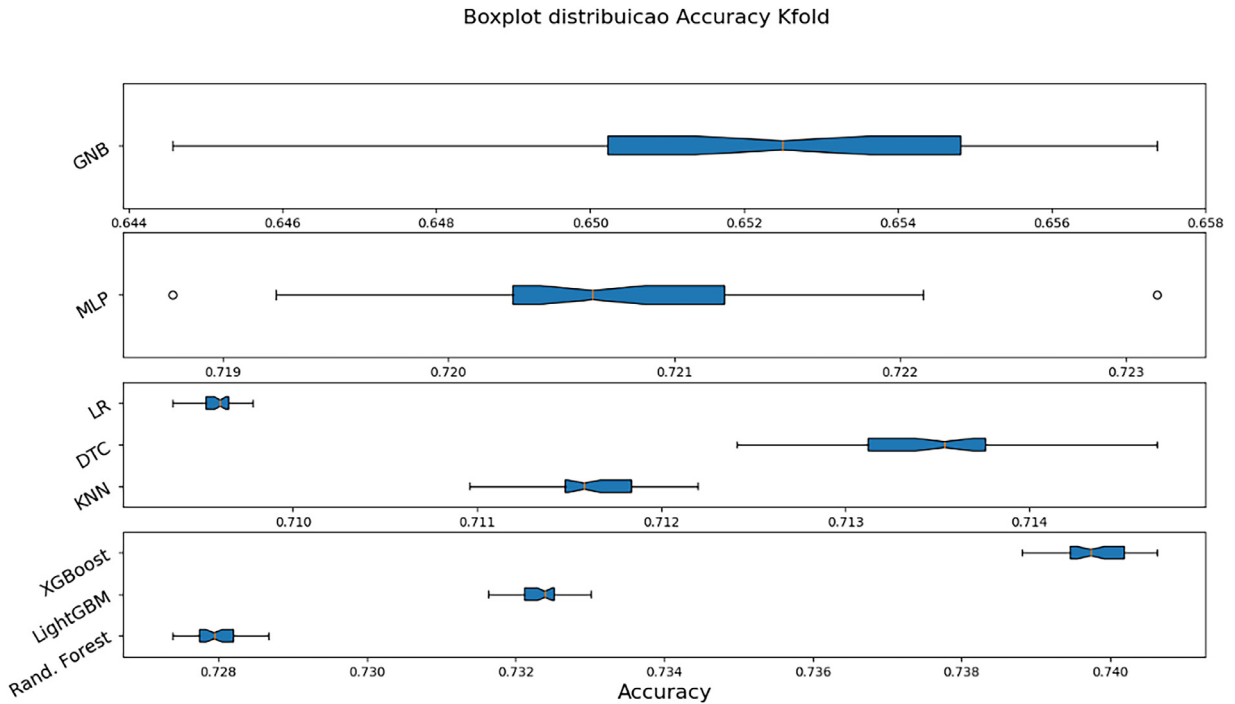


Fig. 8. BoxPlot of the accuracy distribution of Machine Learning models.

Table 6
Dispersion and position measures for KFOLD test results.

Metric	Classifier							
	Naive Bayes	KNN	Decision Tree	Random Forest	LightGBM	XGBoost	Logistic Regression	MLP
mean	0,65243	0,71164	0,71346	0,72797	0,73237	0,73981	0,70959	0,72069
std	0,00325	0,00028	0,00049	0,00035	0,00034	0,00048	0,00010	0,00083
min	0,64457	0,71096	0,71241	0,72738	0,73163	0,73881	0,70934	0,71878
25 %	0,65023	0,71148	0,71312	0,72775	0,73212	0,73946	0,70953	0,72029
50 %	0,6525	0,71158	0,71354	0,72794	0,73239	0,73974	0,70960	0,72064
75 %	0,65482	0,71184	0,71376	0,72820	0,73252	0,74019	0,70965	0,72122
max	0,65736	0,7122	0,71469	0,72868	0,73301	0,74063	0,70978	0,72314

standing out, both with percentages above 76 %. Regarding overall performance, measured by the accuracy metric, all algorithms offer results above 59 %, with XGBoost and LightGBM standing out, achieving an accuracy of 73 %. Next, we have Random Forest with 72 %, and Decision Tree and Logistic Regression with 71 %. Therefore, of the eight classifiers, five have accuracy above 70 %, indicating the robustness of the proposed model. Additionally, the results suggest significant learning, surpassing the randomness of the phenomenon.

After testing with the confusion matrix, the algorithms were evaluated using the KFOLD technique, configured with ten divisions for the dataset and 40 iterations over the KFOLD test. This procedure allowed us to obtain the accuracy distribution for each of the machine learning models, visualized through the following Box Plot in Fig. 8.

The boxplot graphs presented above allow visually identifying the quartiles, upper and lower limits, as well as the outlier points associated with each of the accuracy distributions of the algorithms. It is observed that only the MLP algorithm presented outlier points, while the XGBoost algorithm showed the highest interquartile range, whereas the other algorithms showed smaller interquartile distances.

On the other hand, the results of the KFOLD tests can be analytically visualized, as shown in Table 6, where classical dispersion and position metrics are presented, as well as the quartiles (represented by 25 %, 50 %, and 75 %) and maximum values for the accuracy distribution of each of the classifier algorithms.

It is observed that seven out of eight classifiers present an accuracy average ranging from 71 % to 73 %, with low variance. These numbers indicate significant generalization capability and high precision of these algorithms. Therefore, the results show that the classification algorithm models proposed by this work are accurate and provide decision-making with an average accuracy of 70 %, supported by seven algorithms, providing a high degree of reliability.

After a detailed analysis of crashes involving truck drivers on Brazilian federal highways between 2013 and 2022, significant patterns have been identified. Initially, there was a deceleration in fatal incidents between 2013 and 2016, followed by a phase of oscillations with an upward trend from 2017. The investigation of the most dangerous highways, the role of road type in crash occurrence, and the main causes of crashes provide valuable insights into the challenges faced in road safety. The influence of external factors, such as weather conditions, is also evident. The application of machine learning models reveals a significant capacity for prediction and decisive decision-making. Together, these analyses provide a comprehensive understanding of the landscape of road crashes, essential for guiding efforts in prevention and risk mitigation.

Conclusion

The comprehensive analysis of crashes involving truck drivers on Brazilian federal highways from 2013 to 2023 has provided invaluable insights into the dynamic behavior of these incidents. By examining historical mortality data, critical years, and months with the highest incidence of fatal crashes have been pinpointed, shedding light on significant trends and patterns.

The mapped data highlight the pressing need for collaborative efforts between states and the federal government to enhance road conditions, particularly in Minas Gerais, which accounts for half of the country's ten most hazardous highways, followed by Paraná with three. While São Paulo and Rio de Janeiro have fewer highways among the top ten, they underscore the issue's significance. The observed deceleration in fatal incidents from 2013 to 2016, followed by an upward trend from 2017, with annual fluctuations, emphasizes the dynamic nature of road safety challenges, necessitating adaptive and proactive measures.

The analysis of highways with the highest death tolls highlighted MG-381 as the deadliest, followed by other important routes in Minas Gerais, Paraná, Rio de Janeiro, and São Paulo. When investigating the type of track, it was found that highways with single lanes are the most prone to crashes, while those with double and multiple lanes exhibit greater safety.

The main causes of crashes, including vehicle collisions and cargo rollovers, underscore the importance of prudent driving behavior and preventive measures. The necessity to increase the number of lanes on single-lane highways, particularly focusing on areas with the highest fatality rates, is evident. Conversely, highways with more lanes tend to experience higher numbers of minor injuries. Preventive action is crucial in addressing the root causes of crashes. This entails educating drivers about maintaining a safe distance from other vehicles, enacting legislation mandating additional safety features in trucks like blind spot sensors and minimum distance alarms. Moreover, it is imperative to expand priority lanes for trucks, improve stability controls for cargo trailers, bolster cargo inspection procedures to ensure compliance with vehicle weight limits, enforce speed limits rigorously, and consider legislation to mitigate the impact of exhaustive shifts on drivers' working conditions.

The use of machine learning models, which achieved an accuracy exceeding 70 % in predicting accident occurrences, represents a significant additional tool in road safety management, with the XGBoost and LightGBM models standing out by achieving a predictive accuracy of 73 %. By employing the KFOLD strategy, these models demonstrate adaptability and reliability, providing a practical tool to predict accidents on highways proactively. This predictive capability enables policymakers and stakeholders to implement targeted interventions and proactive measures, thus improving overall road safety and potentially saving countless lives.

The work presented in this paper is limited to 5 states in Brazil and decade long traffic data obtained from the BAT of the PRF. An assessment of traffic data from all the states [if available] and over a longer period might put the present findings in perspective and assess if the traffic accident trends and, hence, the modelling and recommended policies are applicable nationwide. Future research could delve into the potential integration of emerging technologies, such as autonomous vehicles and intelligent transportation systems, to further enhance road safety. These studies could investigate how advanced driver assistance systems, vehicle-to-vehicle communication technologies, and predictive analytics can be leveraged to mitigate accident risks and improve overall traffic safety on Brazilian highways. This forward-looking approach is based on the insights gained from the comprehensive analysis conducted in this study, providing a pathway to proactively address road safety challenges and continue advancing the effectiveness of preventive measures and interventions.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Rodrigo Duarte Soliani: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ana Rita Tiradentes Terra Argoud:** Writing – original draft, Methodology, Formal analysis. **Fábio Santiago:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alisson Vinicius Brito Lopes:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Nwabueze Emekwuru:** Writing – review & editing, Visualization, Validation, Formal analysis.

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Further reading

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